

Null-Calibrated Invariant Lag Graphs for Robust and Interpretable Energy Forecasting

Cai Wenjie
Xiamen University
19020232202343@stu.xmu.edu.cn

Abstract

Energy forecasts often depend on many lagged load, weather and calendar variables, but these models can conceal which relations remain useful as operating conditions change. Sparse lag selection is attractive for interpretability, yet repeated selection from an autocorrelated series does not establish that a predictor is more credible than a temporally structured null. Here we introduce null-calibrated invariant lag graphs (NCILG), a conservative screening and forecasting pipeline. NCILG evaluates lag relations across chronological environments using moving-block resampling, compares real predictors with circularly shifted negative controls that preserve marginal distributions and approximate autocorrelation, and retains only relations that exceed the resulting null score distribution. In 100 known-graph systems spanning linear Gaussian, nonlinear heavy-tailed, regime-shift and weak-edge settings, NCILG achieved aggregate F1 of 0.940, precision of 1.000, zero false additions and mean structural Hamming distance of 1.09. On appliance-energy forecasting, strict graph selection reduced 112 features to 16 and graph-restricted HistGradientBoosting obtained MAE of 42.43, 47.34 and 46.52 at 1, 6 and 24 hours. On four years of household electricity data, the graph reduced a Ridge design from 39 to 24 features with less than 1% average MAE change, although unrestricted nonlinear models remained stronger at several horizons. NCILG therefore supports a robustness–interpretability trade-off for forecasting. The real-data edges are stable predictive lag relations, not identified causal effects.

1 Introduction

Energy systems are observed through many partially redundant signals. Forecasts may draw on recent load, weather, calendar variables and multiple lags of each measurement. This flexibility can improve predictive accuracy, but it creates high-dimensional models whose selected inputs are difficult to interpret and can be sensitive to the training interval. The central problem is therefore not only to select fewer variables, but to identify lagged relations that remain useful when the operating regime changes.

Sparse autoregression and Granger-style screening provide transparent summaries of lagged predictability [13, 4]. Stability selection asks whether a candidate is repeatedly selected after perturbing the sample [9]. However, a fixed selection frequency has no direct temporal baseline. Independent resampling or random permutation erases local serial dependence and can make a null predictor unrealistically weak. A candidate that is frequently selected may therefore still be no more informative than a predictor with strong autocorrelation but no meaningful timing relative to the target.

The distinction between predictive structure and causality is equally important. PCMCI and PCMCI+ estimate conditional lagged and contemporaneous time-series relations [12, 11], whereas VAR-LiNGAM seeks a structural vector autoregression under non-Gaussian assumptions [6]. Invariant prediction can link stable mechanisms to causal conclusions when its environmental and mechanism assumptions are justified [10]. Those assumptions are not established by observational household electricity records. We therefore restrict the present claim to stable predictive lag relations.

We introduce NCILG, a conservative graph-screening and forecasting procedure. The training period is divided into chronological environments. Moving-block bootstrap samples preserve local serial dependence within each environment [7]. For every observed predictor, a circularly shifted copy supplies a negative control with the same marginal distribution and approximate autocorrelation but disrupted cross-variable timing. A candidate edge must exceed the empirical null distribution, retain a stable coefficient sign and remain supported across environments. Stable parents then define a graph-restricted residual forecaster that is blended, on a past validation window, with a full-feature forecast to limit the cost of an overly sparse graph.

The study contributes an autocorrelation-aware negative-control calibration for lag screening; a conservative cross-environment score; a graph-restricted forecaster with a validation-risk blend; and deliberately separated evaluations of structural recovery and real-data forecasting. The last separation is essential: simulations can assess recovery because their graph is known, while real-data figures visualize predictive structure rather than identified causal effects.

2 Background and scope

Lasso regression [13] is widely used for sparse autoregression and Granger screening. Stability selection repeatedly applies a base selector to perturbed samples and retains frequently selected variables [9]. In time series, independent resampling is inappropriate because it erases local dependence; moving-block bootstrap preserves local stationary dependence [7]. The present method uses these tools for conservative screening rather than as evidence of a causal graph.

Invariant causal prediction tests whether conditional mechanisms remain stable across environments [10]. Our score borrows the principle of invariance but is not an implementation of invariant causal prediction and does not inherit its identification theorem. Negative controls are used to expose residual bias [8]; here, circular shifts adapt that logic to autocorrelated time series. When random variables are present in a dataset, they provide an additional empirical check rather than an unqualified causal control.

Forecasting baselines range from persistence and linear autoregression to tree ensembles and neural sequence models. DLinear demonstrates that simple temporal linear models can be highly competitive on long-horizon benchmarks [14]. For graph recovery we compare not only pooled stability but Lasso-VAR, PCMCI, PCMCI+, and VAR-LiNGAM under identical observations and lag limits. For forecasting we include linear, nonlinear, and neural baselines so that the graph method does not benefit from a weak comparison class.

3 Method

3.1 Lagged regression in chronological environments

Let $x_t \in \mathbb{R}^p$ and let $\mathcal{L}$ be a set of positive lags. For target j , the base selector fits

$$\hat{\beta}_j = \arg \min_{\beta} \frac{1}{2n} \left\| x_{j,t} - \sum_{\ell \in \mathcal{L}} \beta_{j,\ell}^\top x_{t-\ell} \right\|_2^2 + \lambda \|\beta\|_1. \quad (1)$$

The training sequence is divided into E contiguous environments. Within environment e , B moving-block bootstrap samples yield selection indicators $I_{b,e,i \rightarrow j,\ell}$ and coefficient signs.

For count $k_{e,i \rightarrow j,\ell} = \sum_b I_{b,e,i \rightarrow j,\ell}$, the Wilson lower confidence bound is

$$L_e = \frac{\hat{p} + z^2/(2B) - z\sqrt{\hat{p}(1-\hat{p})/B + z^2/(4B^2)}}{1 + z^2/B}, \quad (2)$$

Algorithm 1 Null-calibrated invariant lag graph

Require: multivariate training series, lags $\mathcal{L}$, environments E

- 1: **for** each chronological environment **do**
 - 2: construct circular-shift negative controls
 - 3: **for** $b = 1, \dots, B$ **do**
 - 4: sample moving blocks and fit sparse lag models
 - 5: record edge inclusion and coefficient sign
 - 6: **end for**
 - 7: compute Wilson lower scores
 - 8: **end for**
 - 9: aggregate real scores using Eq. (4)
 - 10: calibrate the threshold from pooled environment-level null scores
 - 11: fit full and graph residual forecasters
 - 12: select the robust convex blend on a past validation window
-

where $\hat{p} = k/B$. The environment edge score is

$$S_{e,i \rightarrow j, \ell} = L_e \left| B^{-1} \sum_b \text{sign}(\hat{\beta}_{b,e,i \rightarrow j, \ell}) \right|. \quad (3)$$

The second factor suppresses coefficients that are frequently nonzero but reverse sign.

3.2 Autocorrelation-preserving null calibration

For each observed variable, we construct a negative control by a circular shift larger than two bootstrap blocks. The transformation preserves the marginal distribution and most within-series autocorrelation, while breaking the original alignment with the remaining variables. Real and null lagged predictors are entered into the same Lasso fit, so they face the same target, penalty, and resampling path.

The invariant score uses a lower environment quantile,

$$S_{i \rightarrow j, \ell}^{\text{inv}} = Q_{0.25}(\{S_{e,i \rightarrow j, \ell}\}_{e=1}^E). \quad (4)$$

Let $\mathcal{N}$ be the set of scores for shifted controls. The threshold $q_{1-\alpha}(\mathcal{N})$ is an empirical high null quantile. If null and noncausal scores are exchangeable, the rank construction provides finite-sample calibration for a new null score. Because temporal exchangeability is only approximate, we report the achieved control exceedance and treat this as calibration, not formal false-discovery-rate control.

3.3 Graph-restricted residual forecasting

For forecast horizon h , persistence gives $\tilde{y}_{t+h} = y_t$. Full and graph models predict the residual $r_{t+h} = y_{t+h} - y_t$ from either all lagged variables or the selected parents plus target lags and calendar features. Let $\hat{y}^F$ and $\hat{y}^G$ denote the two forecasts. A past validation window selects

$$\hat{w} = \arg \min_{w \in [0,1]} \overline{|y - [(1-w)\hat{y}^F + w\hat{y}^G]|} + \gamma Q_{0.90}(|y - [(1-w)\hat{y}^F + w\hat{y}^G]|). \quad (5)$$

The deployed forecast is $(1 - \hat{w})\hat{y}^F + \hat{w}\hat{y}^G$. This “safe blend” is a validation-risk safeguard; it is not a distribution-free test-time guarantee.

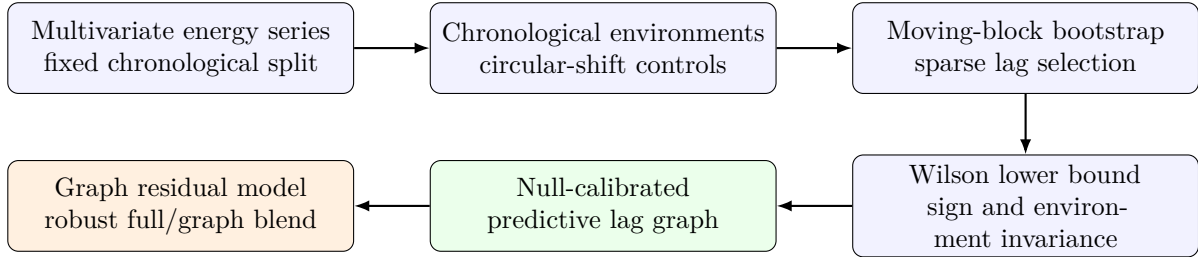


Figure 1: NCILG separates graph calibration from forecast fitting. Real-data edges are interpreted as stable predictive lags.

4 Experimental Design

4.1 Known-truth simulations

We generate 100 independently seeded, eight-variable systems (25 per scenario), each with invariant edges at lags one and two and four chronological regimes. The scenarios are linear Gaussian noise, nonlinear dynamics with Student- t innovations, regime-dependent mechanism scaling, and weak invariant edges. Every system also contains additional links active in only one regime and excluded from invariant ground truth. Each series has 3,500 retained observations. We compare NCILG with ordinary pooled stability, Lasso-VAR, PCMCI, PCMCI+, and VAR-LiNGAM under the same maximum lag of two. Hyperparameters are fixed across all systems; no oracle edge count or test labels are used. We report precision, recall, F1, structural Hamming distance (SHD), and false edges. This design follows the broader principle that causal-discovery evaluation needs known graphs rather than qualitative real-data plausibility [3].

4.2 Real energy datasets

UCI Appliances Energy Prediction contains 19,735 ten-minute observations from a low-energy house, including appliance demand, indoor conditions, weather, and two explicitly random variables [2, 1]. Those random variables provide genuine negative controls in addition to circular shifts. Forecast horizons are 1, 6, and 24 hours.

UCI Individual Household Electric Power Consumption contains approximately four years of one-minute measurements [5]. We aggregate to hourly means, producing 34,190 complete observations of active/reactive power, voltage, intensity, and three sub-metered loads. This dataset has fewer variables but a longer temporal span.

Both datasets use five expanding rolling origins. Baselines are persistence, full Ridge, full HistGradientBoosting, graph-restricted Ridge/HistGradientBoosting, DLinear, and a two-layer GRU. Neural models use 24 hours of context on Appliances and seven days on the hourly household series, AdamW optimization for at most 35 epochs, and past-only early stopping. Hyperparameters and blend weights use only past training/validation observations.

4.3 Implementation details

Real-data graph estimation uses four chronological environments and 36 moving-block replicates per environment. Appliance lag indices are (1, 3, 6) ten-minute steps, with Lasso penalty 0.018 and block length 144; household lags are 1, 6, and 24 hours, with penalty 0.014 and 24-hour blocks. The real-data deployment point uses a 95% Wilson lower bound and the 99.5th null percentile. The simulation benchmark uses 24 replicates, an 80% lower bound ($z = 1.28$), and the 99th null percentile, fixed before the 100-system run to reduce weak-edge under-selection. These are two declared operating points

Table 1: Known-truth graph recovery over 100 systems (25 per scenario).

Method	Prec.	Recall	F1	SHD	False
Lasso-VAR	0.881	0.511	0.600	5.49	0.11
PCMCI	0.729	0.999	0.842	4.16	4.15
PCMCI+	0.662	1.000	0.795	5.73	5.73
VAR-LiNGAM	0.786	0.996	0.878	3.03	2.99
Ordinary stability	0.824	0.997	0.899	2.61	2.58
NCILG	1.000	0.901	0.940	1.09	0.00

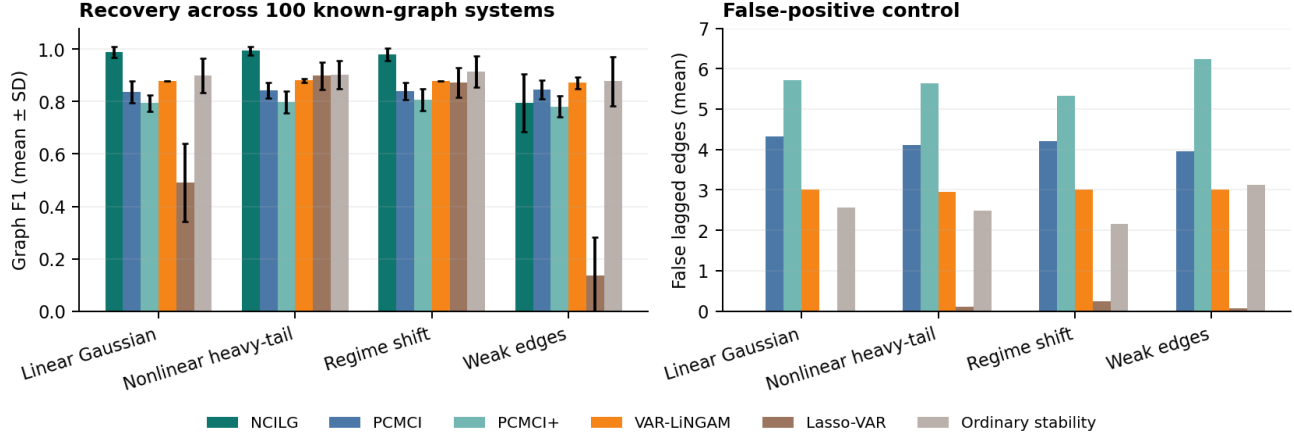


Figure 2: Known-truth recovery across four generators. Error bars are between-system standard deviations. Null calibration removes regime-specific false positives; the cost appears as lower recall under deliberately weak invariant edges.

rather than per-scenario tuning. Forecast designs include current value, one-step, one-hour, six-hour, daily, and weekly lags when available, plus time-of-day and day-of-week harmonics. Every rolling origin fits preprocessing on its own past. Neural models use batch size 192, AdamW learning rate 8×10^{-4} , and early-stopping patience six.

5 Results

5.1 Graph recovery

Table 1 aggregates all 100 systems. NCILG has the highest mean F1, perfect precision, and no false additions. PCMCI, PCMCI+, and VAR-LiNGAM attain near-perfect recall but retain three to six regime-specific links per system. Lasso-VAR is conservative but misses roughly half of the invariant graph. NCILG’s residual error is concentrated in the deliberately weak-edge scenario: F1 is 0.989, 0.994, 0.981, and 0.795 across the four scenarios, respectively. This exposes the intended trade-off rather than hiding it in a single favorable generator.

5.2 Appliance forecasting

On Appliances, strict null thresholding retains the ten-minute lights edge together with the target’s own stable ten- and 60-minute lags (Fig. 3). The forecast design includes those relations and calendar terms, reducing 112 features to 16 without rank-based feature padding. Table 2 shows that graph Ridge

Appliances: null-calibrated predictive lag graph

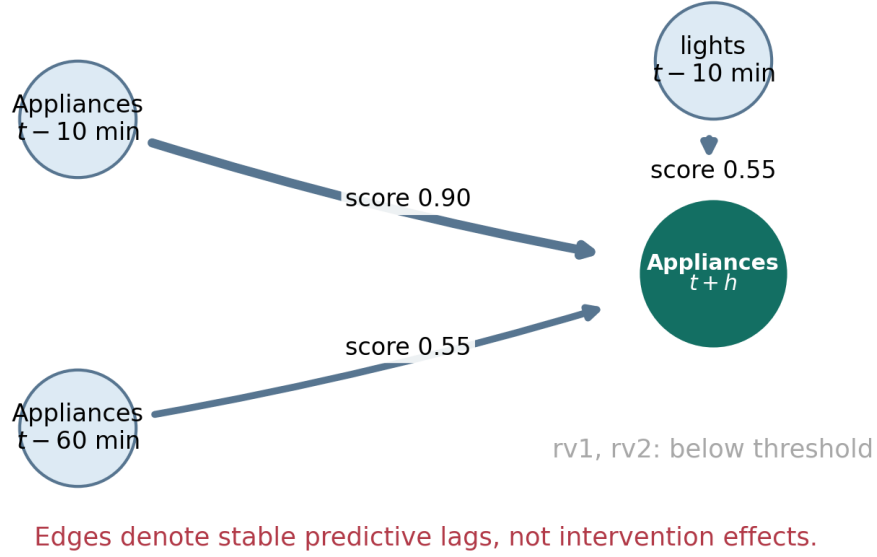


Figure 3: Strictly thresholded Appliances lag graph. Nodes are time-unrolled for display; edge labels are invariant scores. Random controls remain below threshold.

improves on full Ridge at every horizon. Graph-restricted HistGradientBoosting is strongest at all three horizons, reaching MAE 42.43, 47.34, and 46.52. Its MAE is lower than full Ridge at every one of the five rolling origins for each horizon, rather than being driven by a single favorable split. The safe linear blend remains competitive at 43.10, 48.30, and 47.42. DLinear and GRU use all observed inputs; neither dominates the strict graph predictor on this dataset.

5.3 Long-span household forecasting

The household graph retains Voltage, Sub-metering 1, and Sub-metering 2, plus the target’s own lags and calendar terms. The safe blend changes MAE relative to full Ridge by -0.33% , $+0.51\%$, and $+0.40\%$ across the three horizons while using 38% fewer features. A full HistGradientBoosting model is substantially better at one hour (0.308 MAE) and 24 hours (0.442), while the GRU is strongest at six hours (0.451). Thus NCILG does not claim the best unconstrained forecast. Its value is a compact linear forecast whose selected inputs persist across time regimes. Figure 5 makes the compression cost explicit instead of presenting sparsity as a free improvement.

6 Discussion

The expanded synthetic benchmark isolates the role of invariance across distributional conditions. Pooled stability and modern time-series discovery methods accept edges that are genuinely predictive in part of the sequence but absent elsewhere. Lower environment quantiles reject these edges without sacrificing aggregate F1, and negative controls make the threshold data-dependent. The weak-edge cell is the exception: perfect precision is purchased with lower recall. Applications prioritizing discovery rather than screening should therefore use a less conservative operating point or report a ranked graph.

On real data, graph restriction is most useful when the full design contains many related temperature and humidity lags. The strict appliance graph is exceptionally sparse yet supports both linear and

Table 2: Rolling-origin MAE on real energy datasets. Lower is better.

Dataset	Method	Features	1 h	6 h	24 h
Appliances	Persistence	0	49.02	68.41	52.28
	Full Ridge	112	47.92	53.99	53.24
	Graph Ridge	16	43.65	49.17	46.75
	NCILG safe blend	16	43.10	48.30	47.42
	Graph HistGB	16	42.43	47.34	46.52
Household	Persistence	0	0.380	0.793	0.541
	Full Ridge	39	0.371	0.527	0.475
	Graph Ridge	24	0.376	0.540	0.480
	NCILG safe blend	24	0.370	0.529	0.477

Table 3: Five-origin MAE for neural forecasting baselines.

Dataset	Method	1 h	6 h	24 h
Appliances	DLinear	51.15	52.93	52.42
	GRU	46.90	48.84	48.66
Household	DLinear	0.431	0.459	0.481
	GRU	0.425	0.451	0.463

nonlinear forecasters. The smaller household system offers less redundancy, so compression yields little accuracy gain. No single architecture dominates both datasets: graph-restricted HistGradientBoosting leads all Appliances horizons, the GRU leads the six-hour household task, and full HistGradientBoosting leads household one- and 24-hour forecasting.

Several limitations are fundamental. Circular shifts only approximate null exchangeability and can be invalid under strong seasonality shared across all offsets. On the Household data, discrete selection-score ties cause 9.2% of null scores to equal or exceed the nominal 99.5th-percentile threshold, illustrating that a nominal quantile need not imply exact finite-sample false-selection control. Chronological segments are observational environments, not randomized interventions. Instantaneous effects, latent confounding, measurement aggregation, and feedback remain unresolved. Therefore real edges are predictive lag relations. Structural causal language is reserved for the simulations, where the graph is known.

7 Conclusion

NCILG combines autocorrelation-aware negative controls, environment invariance, and robust forecast blending. Across 100 known-graph systems it outperforms four established lag-discovery baselines in aggregate F1 while adding no false edges, with a clearly identified weak-edge recall limit. It also improves a sparse residual forecast on appliance data. On a longer household dataset it preserves full-Ridge accuracy with a smaller feature set but does not beat a full nonlinear model. These results define a useful and honest scope: robust, interpretable lag screening for energy forecasting rather than causal identification from observational data alone.

Data and Code Availability

Both real datasets are publicly available from the UCI Machine Learning Repository under CC BY 4.0 licenses. Scripts store graph scores, null thresholds, selected parents, rolling-origin metrics, random

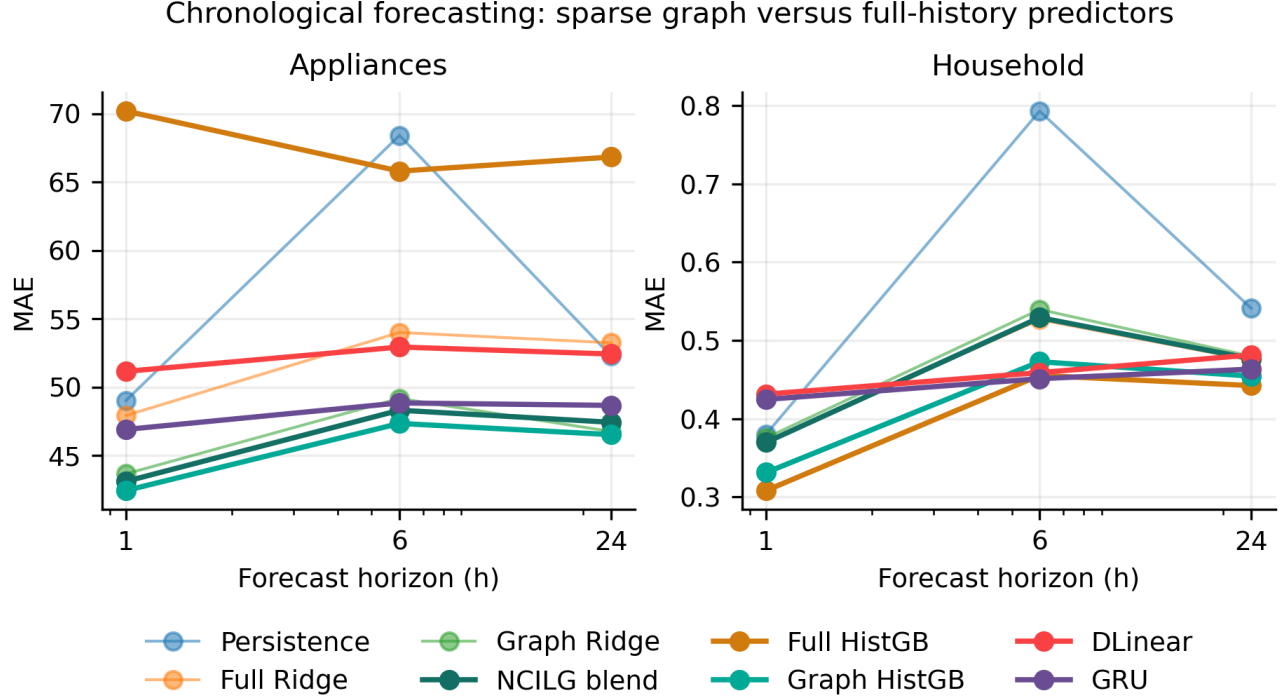


Figure 4: Five-origin chronological forecasting. Graph restriction improves the high-dimensional appliance problem; the unrestricted nonlinear model remains strongest on the longer household series.

seeds, and all ablations.

References

- [1] Luis Candanedo. Appliances energy prediction, 2017.
- [2] Luis M. Candanedo, Véronique Feldheim, and Dominique Deramaix. Data driven prediction models of energy use of appliances in a low-energy house. *Energy and Buildings*, 140:81–97, 2017.
- [3] Yuxiao Cheng, Ziqian Wang, Tingxiong Xiao, Qin Zhong, Jinli Suo, and Kunlun He. Causalttime: Realistically generated time-series for benchmarking of causal discovery. In *International Conference on Learning Representations*, 2024.
- [4] Clive W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969.
- [5] Georges Hebrail and Alice Berard. Individual household electric power consumption, 2006.
- [6] Aapo Hyvärinen, Kun Zhang, Shohei Shimizu, and Patrik O. Hoyer. Estimation of a structural vector autoregression model using non-gaussianity. *Journal of Machine Learning Research*, 11(56):1709–1731, 2010.
- [7] Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.

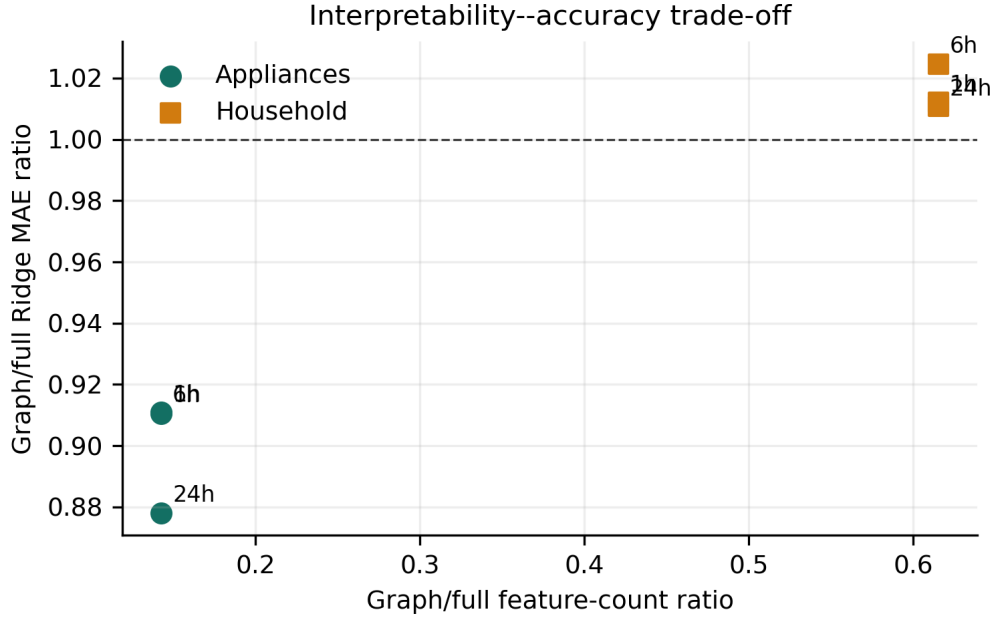


Figure 5: Feature compression versus Ridge MAE. Points below the dashed line improve accuracy while using fewer lagged predictors; points above it quantify the price of interpretability.

- [8] Marc Lipsitch, Eric Tchetgen Tchetgen, and Ted Cohen. Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388, 2010.
- [9] Nicolai Meinshausen and Peter Bühlmann. Stability selection. *Journal of the Royal Statistical Society: Series B*, 72(4):417–473, 2010.
- [10] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78(5):947–1012, 2016.
- [11] Jakob Runge. Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence*, volume 124 of *Proceedings of Machine Learning Research*, pages 1388–1397, 2020.
- [12] Jakob Runge, Peer Nowack, Marlene Kretschmer, Seth Flaxman, and Dino Sejdinovic. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, 5(11):eaau4996, 2019.
- [13] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.
- [14] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023.